

Operational Resilience Enhancement of Water-Energy Microgrids via ReserveMax Nonlinear Model Predictive Control

Saskia Putri¹, Faegheh Moazeni^{1,*}, Javad Khazaei²

⁽¹⁾ *Dep. of Civil and Environmental Engineering, ⁽²⁾ Dep. of Electrical and Computer Engineering*
Lehigh University, Bethlehem PA, USA

Emails: sap322@lehigh.edu, moazeni@lehigh.edu, khazaei@lehigh.edu

Abstract—This paper presents a formulation of simultaneous control and model for an integrated water-energy microgrid system, enhancing the power reserve through optimal pump scheduling and power dispatch. Leveraging the predictive capability of nonlinear model predictive control (NMPC), the closed-loop control framework maximizes power reserve in a safe and cost-effective manner for water distribution systems (WDS) and grid-tied microgrids (MG). To evaluate system resilience, two performance metrics are introduced: the Energy Reserve Index (ERI), which quantifies the system's ability to meet critical power demands, and the Stored Resource Availability Index (SRAD), which assesses long-term resource sufficiency by measuring stored water and energy reserves relative to safety thresholds. Despite seasonal variations in water demand and renewable generation, the proposed NMPC ensures optimal pump flow rates and power dispatch, consistently meeting water and power demands while satisfying operational constraints and power reserves. Compared to an economic dispatch open-loop strategy, the proposed control achieves 20% cost reduction while simultaneously increasing energy and water reserves by 30% and 20%, respectively, demonstrating significant improvements in system performance and resilience.

Index Terms—Water-energy nexus, water distribution system, grid-connected microgrid, power reserve, demand response.

I. INTRODUCTION

Growing attention has been paid over the years to co-optimizing water distribution systems (WDS) and power systems. By coordinating water and power operations, these systems can better manage resources to meet fluctuating demand, mitigate shortages, and optimize energy usage in response to varying conditions [1]. This integrated approach is particularly beneficial for energy-intensive water system components, as it reduces operational costs and improve sustainability by synchronizing water flow with energy availability [2]. Recent studies have highlighted the benefits of coordinated operation between these systems for improved efficiency and robustness [3], [4]. Due to the aging electricity network, the application of microgrids (MGs) has emerged, offering an effective solution for integrating renewable energy sources (RES) and maintaining a reliable power supply for WDS [5], [6].

Various studies have explored different optimization techniques for managing an integrated water-energy MG, including day-ahead operation with five demand-response model [7], effective management to multiple generation units [6] and

design and operation centered on total cost minimization [8]. However, many existing optimization frameworks employ an open-loop approach, which assumes a static solution without real-time feedback.

In response, Model Predictive Control (MPC) addresses a series of open-loop control problems using receding horizon control (RHC), where it executes the first control action and continuously shifts the prediction horizon forward. This approach enables simultaneous feedback, feedforward, and optimal control, making MPC an advanced and favored control strategy [9]. Our previous work in [10] has explored the effectiveness of MPC in ensuring the economic dispatch of the integrated water-energy MG by leveraging the energy flexibility from the WDS. Despite its advantages, the application of MPC remains limited, with most studies focusing on WDS and power system operations individually. This highlights a research gap for a unified MPC framework tailored to challenges in an integrated water-energy microgrid.

Power reserve management is another critical aspect of optimizing the integrated operation of water-energy MG, as it ensures that sufficient energy is available to meet sudden demand spikes or renewable supply fluctuations. Although previous studies have recognized the importance of power reserves [11], [12] for stand-alone MGs, there remains a gap in the literature on optimal reserve management, specifically within integrated water-energy MGs. A common objective of optimization algorithms in MG is to minimize costs and balance loads, but power reserves are not explicitly considered.

This study addresses these gaps by proposing a novel nonlinear MPC, named *ReserveMAX* NMPC, specifically designed for power reserve optimization in an integrated water-energy MG. The main contributions are summarized as follows

- 1) Coordinated operation of the WDS and grid-tied microgrid is proposed. The proposed integrated system modeling promotes maintaining power reserve by introducing operative reserve constraints to ensure reliable operation in the presence of fluctuating water and power demands, as well as uncertain RES power generation.
- 2) A closed-loop control scheme of NMPC tailored for maintaining reserves and cost-efficient operation in an integrated water-energy microgrid is developed.
- 3) In contrast to conventional methods that rely on two-stage optimization, this work applies a simultaneous nonlinear control strategy, allowing for coordinated real-time management of the integrated water-energy microgrid.

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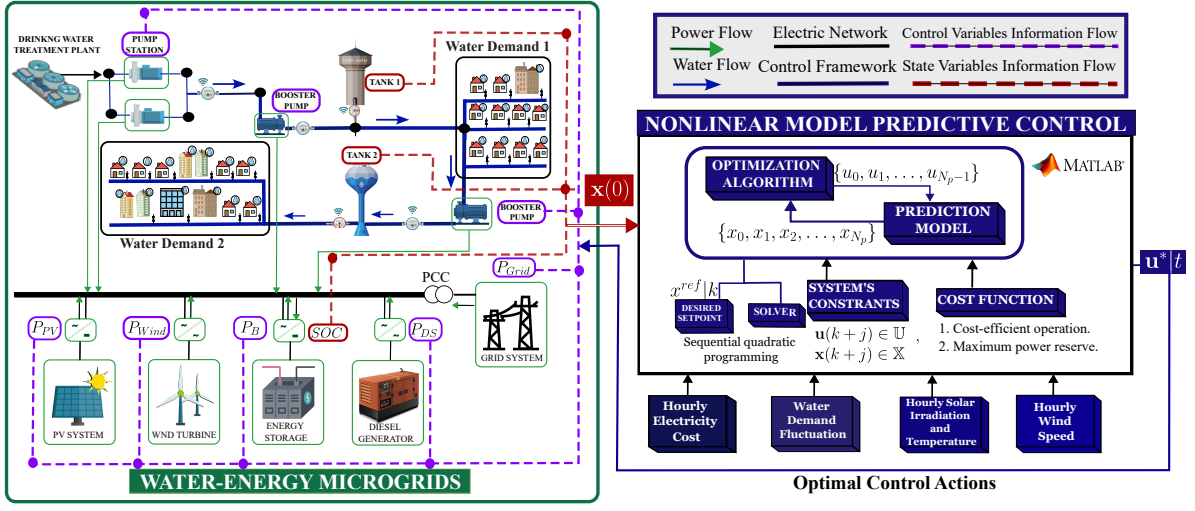


Fig. 1: Proposed control scheme of integrated water-energy microgrid

- 4) Novel performance metrics are introduced for an integrated water-energy microgrid to investigate the system's reliability and stability.

The remainder of the paper is structured in the following order. Section II describes the structure of the proposed control strategy and integrated water-energy microgrid. Section III presents mathematical model formulation of both systems. Details on ReserveMAX NMPC formulation are described in Section IV. Section V presents the case study description analysis, and Section VI concludes the paper.

II. WATER-ENERGY MICROGRIDS: DESCRIPTION OF PROPOSED FRAMEWORK

The structure of the proposed control system for an integrated water-energy MG is illustrated in Fig. 1. The studied integrated system supports two aggregated water demand zones obtained from real water demand in Telford, PA. The WDS includes interconnected components such as pumps and storage tanks distributed across the network while the MG incorporates a battery energy storage system (BESS), diesel generator, wind turbine, and photovoltaic (PV) system. Simultaneous control of the integrated system is proposed, which allows adjustment of the pump schedule to reduce power demand and increase power reserve in the MG to account for RES variability and ensure reliable operation during emergency and critical loads.

III. MATHEMATICAL MODEL FORMULATION OF WATER-ENERGY SYSTEMS

A. System Dynamic Models

System dynamics of the water-energy microgrid components are governed by the rate of change of the stored "energy" in the form of volumetric water for WDS and excess power for the MG, modeled as: follows [9], [13]

$$\frac{dh_{Ti}}{dt} = \frac{1}{A_i} \left(\sum_l q_l^{in}(t) - \sum_j q_j^{out}(t) \right) \quad (1)$$

$$\frac{dSoC}{dt} = \frac{-\eta_b}{C_B} P_B(t) \quad (2)$$

where (1) describes the rate of change in water level within each storage tank with h_{Ti} (in m) and A (in m^2) denote the water levels and constant cross-sectional area of the i -th tank, respectively, q_l^{in} and q_j^{out} represent the l -th inflow and j -th outflow of the i -tank, respectively in m^3/hr . The BESS of the MG is modeled using the state-of-charge (SoC) in (2) with φ_b as the efficiency of the energy conversion during charging (φ_c)

and discharging ($1/\varphi_d$). C_B denote the rated capacity of BESS, and $P_B(t)$ represent the charging and discharging power output of the BESS.

B. Interlink of The Integrated Water-Energy MG

Water-energy microgrids are interconnected through the pumps that maintain the flow and pressure in the water distribution network. In this study, the power demand of the pump is solely met by the grid-connected microgrid. Pump's power consumption ($P_{ui}(t)$) is given by [13]:

$$P_{ui}(t) = \frac{\gamma \cdot r_{ui}(t) \bar{q}_{ui} \cdot H_{p_{ui}}(t)}{\eta_{ui}(t)} \quad (3)$$

where γ is the specific weight of water in kN/m^3 , $\bar{q}_{ui}$ is the maximum flow rate of the pump in m^3/hr , $r_{ui}(t) = \frac{n_m(t)}{n_{nom}}$ is the normalized speed of the pump with $n_m(t)$ and n_{nom} as the measured and nominal speed of the pump in rpm , $H_{p_{ui}}(t) = a_p - b_p(r_{ui}(t) \bar{q}_{ui})^2$ is the pressure head of the pump in m and $\eta_{ui}(t) = \alpha_1(r_{ui}(t) \bar{q}_{ui})^2 + \alpha_2(r_{ui}(t) \bar{q}_{ui}) + \alpha_3$ is the pump's efficiency with coefficients α_i obtained from manufacturers.

C. Power Generation Models

1) *Diesel Generator*: The power output of the diesel generator ($P_{DS}(t)$) is constrained by the power limits, maximum runtime, and minimum downtime requirements. During active operation, the power output is bounded by [11]:

$$\underline{P}_{DS} \cdot v(t) \leq P_{DS}(t) \leq \bar{P}_{DS}(t) \cdot v(t) \quad (4)$$

where $v(t) \in \{0, 1\}$ is a binary variable indicating the generator's state (1 for on and 0 for off) and $\underline{P}_{DS}, \bar{P}_{DS}$ denote the minimum and maximum power output of diesel generator.

To prevent excessive operation, the generator has a maximum runtime $\bar{T}_{DS}$, after which it must remain off until the required minimum downtime $\underline{T}_{DS}$ is met. These constraints are enforced through runtime and downtime counters, $run_{DS}(t)$ and $downt_{DS}(t)$, respectively, updated by:

$$run_{DS}(t+1) = \begin{cases} run_{DS}(t) + 1, & \text{if } v(t) = 1 \\ 0, & \text{if } v(t) = 0 \end{cases} \quad (5)$$

$$downt_{DS}(t+1) = \begin{cases} downt_{DS}(t) + 1, & \text{if } v(t) = 0 \\ 0, & \text{if } v(t) = 1 \end{cases} \quad (6)$$

When $run_{DS}(t)$ reaches $\bar{T}_{DS}$, the generator turns off ($v(t+1) = 0$) and resets run_{DS} . Similarly, if $downt_{DS}(t)$ is less than $\underline{T}_{DS}$, the generator remains off until this condition is satisfied. These constraints ensure compliance with operational limits while maintaining system reliability and fuel efficiency.

2) *Photovoltaic (PV)*: PV power output depends on solar irradiance, system characteristics, and environmental factors, posing significant uncertainty. The PV power output can be modeled as [14]:

$$P_{PV}(t) = P_{PV}^{rated} \frac{G_{ac}(t)}{G_{rated}} (1 + T(T_c - T_i)) \quad (7)$$

where P_{PV}^{rated} (in kW) is the rated power output of PV in standard temperature conditions, $G_{ac}(t)$ and G_{rated} are the actual and rated solar irradiances at time t , T is the temperature coefficient of the PV unit while T_c and T_i are the reference and actual temperature of solar cells in $^{\circ}\text{C}$.

3) *Wind Turbine (WT) Power Generation*: The power output of wind turbines is generated over a wide range of wind speeds to ensure cost-efficient operation and protect against extreme conditions. The power output is defined as [6]:

$$P_{WT}(t) = \begin{cases} 0 & \text{if } v_t < v_i \text{ \& } v_t > v_o \\ P_r \frac{v_t - v_i}{v_r - v_i} & \text{if } v_i \leq v_t \leq v_r \\ P_r & \text{if } v_r \leq v_t \leq v_o \end{cases} \quad (8)$$

where v_t and v_r are the wind speed at time instant t and the rated wind speed, respectively in m/s , P_r is the rated power in kW , v_i and v_o are the cut-in and cut-out wind speed in m/s .

D. Physical and Operational Constraints

To ensure safe operation, all studied WDS and MG components are subject to operational constraints. These constraints are formulated as follows:

$$\underline{h}_{Ti} \leq h_{Ti}(t) \leq \overline{h}_{Ti}, \quad \forall i \in \mathcal{N}_{Tank} \quad (9)$$

$$\underline{r}_{ui} \leq r_{ui}(t) \leq \overline{r}_{ui}, \quad \forall i \in \mathcal{N}_{Pump} \quad (10)$$

$$\underline{\Delta r}_{ui} \leq \Delta r_{ui}(t) \leq \overline{\Delta r}_{ui}, \quad \forall i \in \mathcal{N}_{Pump} \quad (11)$$

$$\underline{SoC} \leq SoC(t) \leq \overline{SoC} \quad (\text{BESS}) \quad (12)$$

$$-\underline{P}_B \leq P_B(t) \leq \overline{P}_B \quad (13)$$

$$\underline{P}_{RES} \leq P_{RES}(t) \leq \overline{P}_{RES}(t) \quad (\text{Renewables}) \quad (14)$$

where $(\underline{\cdot})$ and $(\overline{\cdot})$ denote the lower and upper limits of the tank in (9), pumps' speed in (10), rate of change of pumps' speed in (11), state of charge in (12), and power output for each generation unit in (13)-(14). $\mathcal{N}(\cdot)$ represents WDS components.

E. Integrated Water-Energy MG Flow and Power Balances

1) *Water Flow Balance*: To ensure balanced operation, the total flow entering and exiting each node in the studied WDS must be equal. The water flow balance is given by [13]:

$$q_{d_j}(t) = \sum q_{ij}(t) - \sum q_{ji}(t) \quad (15)$$

where at time t , $q_{d_j}(t)$ is the demand flow (in m^3/hr) of the j -th node, q_{ij} is the flow from node i to node j while q_{ji} is the flow from j to i (in m^3/hr).

2) *Operating Power Reserve Balance*: To ensure a resilient MG operation, the total generated power must satisfy the load requirement from (3) and the minimum operative reserve ($\underline{P}_R$); hence, several reserve constraints are imposed, developed from [11]:

$$P_B(t) + P_{DS}(t) + P_{WT}(t) + P_{PV}(t) + P_{Grid}(t) \geq \sum_i P_{ui}(t) + \underline{P}_R \quad (16a)$$

$$P_R(t) = [\overline{P}_{DS} - P_{DS}(t)] + [\overline{P}_{Bdis} - P_B(t)] + [\overline{P}_{Grid} - P_{Grid}(t)] \quad (16b)$$

$$P_R(t) \geq \sum_i P_{ui}(t) \quad (16c)$$

where P_{Grid} denotes power output from the grid, (16a) guarantees demand and minimum power reserve are satisfied, (16b) defines the amount of reserve power ($P_R(t)$) for every time t , and (16c) ensures that pumping primarily occurs when there is available reserve power in the MG.

F. Resilience Metrics of Water-Energy MG Operation

This study proposes several performance indicators to measure the system's resilience, reliability, and capacity to meet critical power needs.

1) *Energy reserve index (ERI)*: assesses the system's ability to meet critical power demands over a specified time horizon T , expressed as:

$$\text{ERI} = \min \left(\frac{P_{Gen}(t)}{P_{Critical}}, 1 \right) \quad (17)$$

where $P_{Critical}$ is the mean power demand of the pumps while $P_{Gen}(t) = P_{DS}(t) + P_B(t) + P_{WT}(t) + P_{Grid}(t) + P_{PV}(t)$ represents the hourly total power generated from each source.

2) *Stored Resource Availability Index (SRAI)*: assesses the sufficiency of stored resources, such as water in tanks and energy in a BESS, to meet critical demands, defined by:

$$\begin{cases} \text{Water SRAI} &= \frac{1}{T} \sum_{t=1}^T \frac{\min(S(t), RS)}{RS} \\ \text{Energy SRAI} &= \frac{1}{T} \sum_{t=1}^T \frac{\min(E_{ESS}(t), EL)}{EL} \end{cases} \quad (18)$$

where RS is set to ensure the tank level reaches a minimum of 35 psi and EL is the reserve energy threshold set to assure the ESS maintains at least 50% of its energy storage capacity.

While various resilience metrics exist in the literature, this paper focuses on ERI and SRAI due to their importance in assessing both short-term operational resilience and long-term resource sufficiency, ensuring immediate demand fulfillment and sustained reliability in water-energy microgrids.

IV. RESERVEMAX NONLINEAR MODEL PREDICTIVE CONTROL (RESERVEMAX NMPC)

This section briefly describes the closed-loop optimization formulation of the integrated water-energy microgrid. The main objective is maximizing the grid-connected MG's power reserve, thereby ensuring reliable operation during critical load.

A. Prediction Model

Given the system dynamics in (1)-(2) state space representation of the integrated water-energy microgrid is expressed as:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t)) \quad (19)$$

where $\mathbf{x}(t) = [h_{Ti}(t) \ SoC(t)] \in \mathbb{R}^n$ is the state variable, $\mathbf{u}(t) = [q_{ui}(t) \ P_{Gi}(t) \ P_R(t)] \in \mathbb{R}^m$ is the control variable with P_{Gi} denote available power output from source $i = \{DS, B, PV, WT, Grid\}$, $\mathbf{d}(t) = [q_{di}(t) \ \overline{P}_{PV}(t) \ \overline{P}_{WT}(t)] \in \mathbb{R}^q$ is the measured disturbance comprises aggregated demand flow for i th node and maximum available RES power.

B. Objective Function

In this study, the main control objective is maximizing the power reserve $P_R(t)$ while minimizing the operating costs of each generation unit to ensure economic dispatch over a prediction horizon N_p . The proposed ReserveMAX NMPC objective function is given by:

$$- \sum_{k=0}^{N_p-1} \underbrace{\alpha P_R(t+k|t)}_{\ell_R(t+k)} + \sum_{k=0}^{N_p-1} \sum_i \underbrace{\beta_i C_i((P_{Gi}(t+k|t)))}_{\ell_{ED}(t+k)} \quad (20)$$

where α is the weight factor, prioritizing reserve power while β_i is the weight for the cost associated with generation unit $i = \{DS, B, PV, WT, Grid\}$ and $C_i((P_{Gi}(t+k)))$ is the operational cost function for each generation unit i at time $t+k$.

C. ReserveMAX NMPC Problem Formulation

Considering the system modeling and constraints in Section III, prediction model in (19) and objective function in (20), NMPC problem formulation is developed, as follows:

$$\min_{\mathbf{U}|t} - \sum_{k=0}^{N_p-1} \ell_R(t+k) + \sum_{k=0}^{N_p-1} \ell_{ED}(t+k) \quad (21a)$$

$$\text{s.t. } \mathbf{x}(t+k+1|t) = \mathbf{f}(\mathbf{x}(t+k|t), \mathbf{u}(t+k|t), \mathbf{d}(t+k|t)), \quad \mathbf{x}_0 = \mathbf{x}(t) \quad (21b)$$

$$\mathbf{E}\mathbf{u}(t+k|t) + \mathbf{M}\mathbf{d}(t+k|t) = 0 \quad (21c)$$

$$\mathbf{G}\mathbf{u}(t+k|t) + \mathbf{H}\mathbf{d}(t+k|t) \leq \mathbf{b} \quad (21d)$$

$$\mathbf{u}(t+k|t) \in \mathcal{U}, \mathbf{x}(t+k+1|t) \in \mathcal{X} \quad (21e)$$

where $\mathbf{U}|t = [\mathbf{u}(t+k), \dots, \mathbf{u}(t+N_p-1)]$ is the control input sequence over the prediction horizon N_p , $\mathbf{x}_0$ is the state's initial condition obtained from the system at time t . Constraint (21b) represents the prediction model of water-energy MGs dynamics in (19). Constraint (21c) ensures systems balances detailed in (15) and defines reserve power in (16b) while constraint (21d) formulates the power balance (16a) and load generation in WDS (16c) with $\mathbf{E}$, $\mathbf{M}$, $\mathbf{G}$, and $\mathbf{H}$ as system matrices. Constraint (21e) denotes the physical limitation of each water-energy MGs component detailed in (9)-(14).

V. CASE STUDY

This section demonstrates a series of case studies to assess ReserveMax NMPC, highlighting the reliability of the integrated water-energy MG. The studies include: 1) assessing ReserveMax NMPC subject to seasonal variation, 2) comparing ReserveMax NMPC (closed-loop setup) performance against an economic dispatch optimization (open-loop setup). The optimal control problem formulation in (21) is solved using sequential quadratic programming (SQP).

A. Case Study 1: ReserveMAX NMPC Robustness to Weather Variation

This case study assesses the effectiveness of ReserveMAX NMPC, which accounts for fluctuations in water demand and weather variations, representing the renewable energy generation uncertainties across seasons.

Fig. 2 shows the allocation of each component in the WDS and MG to meet water and power demands. The alignment of demand and supply in both systems indicates an effective demand response control strategy despite uncertainties. Peaks in water demand are met by combining both pump and tank flow, offering effective water management strategies as shown in the top subplot. This is enforced as the proposed control strategy aims to maximize the use of stored water, which directly reduces the pump's consumption and impacts higher power reserves in the studied MG while still satisfying the minimum pressure level in the network. The bottom subplot depicts power dispatch across multiple generation sources. During high power demand, ReserveMAX NMPC effectively dispatched all available generation units while satisfying all constraints in (4),(9)-(14). During lower demand periods, the BESS charges and grid reliance decreased, and RES covered most of the load, indicating efficient energy use and cost-saving measures.

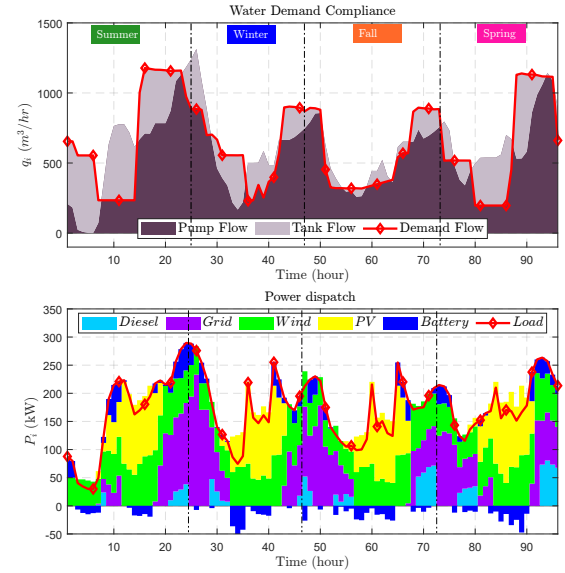


Fig. 2: Case study 1: optimal allocation of water (top subplot) and power resources (bottom subplot) subject to seasonal variation

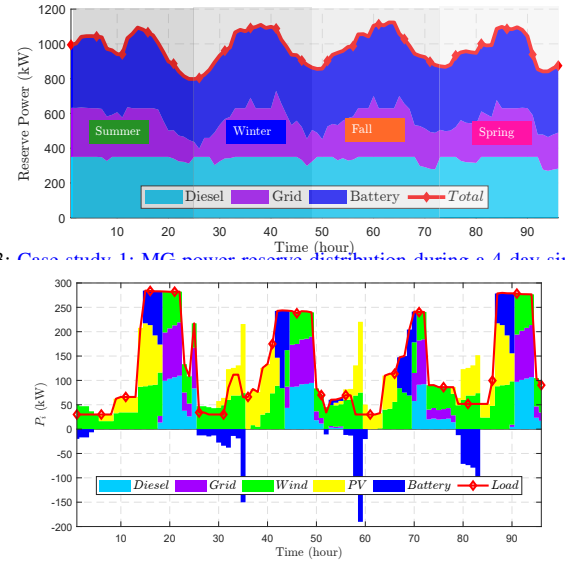


Fig. 3: Case study 1: MG power reserve distribution during a 4-day simulation

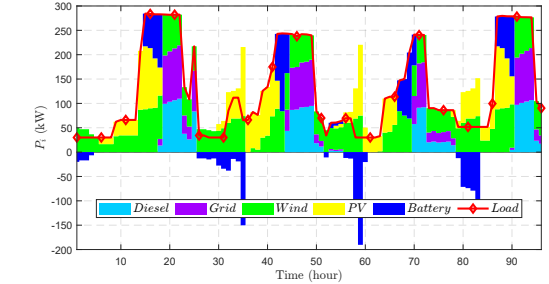


Fig. 4: Case study 2: 96-h economic dispatch of integrated water-energy microgrid (open-loop approach)

Fig. 3 further highlights the power reserve of the studied MG. Power reserves are slightly higher during winter and fall due to reduced water consumption. As observed, diesel and grid power are key reserves that supply power when RES and BESS are insufficient. This aligns with using diesel and the grid as backup sources to stabilize the power supply.

B. Case Study 2: ReserveMAX NMPC (Closed-loop) v. Economic Dispatch Optimization (Open-loop)

This case study is intended to further highlight the effectiveness of ReserveMAX NMPC, leveraging closed-loop formulation to yield optimal control actions, and the relevance of power reserve. The open-loop model formulation followed the system modeling in Section III with problem formulation similar to the well-known optimization algorithm utilized in [15].

Fig. 4 illustrates the power distribution for each generator centered on the economic dispatch of the MG. While power demand is consistently met throughout the simulation, the stored water is excessively used and remains at a minimum level during low water demand, visualized in Fig. 5(a). This impacts pressure stability, requiring sudden surges in pump flow to refill the tank, which may, in turn, affect the lifespan of the pumps. These implications are also observed in the power dispatch where BESS shows significant fluctuations,

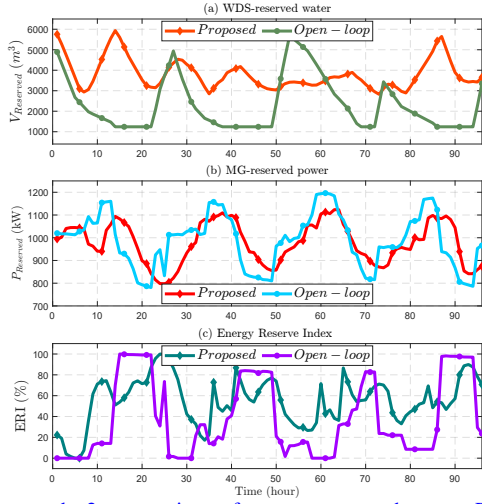


Fig. 5: Case study 2: comparison of energy reserves between ReserveMAX NMPC and an open-loop approach in an integrated water-energy microgrid.

TABLE I: Quantification Performance of Integrated water-energy MG operation

Metric	Improvement (%)
MG operation cost (\$/d)	23.69%
Average Energy Reserve Index (ERI)	33.88%
Energy SRAI	15.74%
Water SRAI	21.64%

with frequent charging and discharging cycles. In addition, the irregular use of diesel indicates that the open-loop system does not prioritize minimizing diesel fuel consumption. This is suboptimal, as diesel is typically a costly and environmentally impactful energy source.

Fig. 5 displays the comparison between the proposed control strategy (ReserveMAX NMPC) with the open-loop formulation based on water (Fig 5(a)) and power reserve (Fig 5(b)) as well as reliability of the MG to meet critical load (Fig 5(c)). As shown in Fig 5(a)-(b), the proposed control strategy managed to maximize the power reserve without sacrificing the water reserve. Meanwhile, the open-loop approach dropped the reserved water volume significantly, making the system vulnerable to water shortages during peak demand. A similar trend is identified for the ERI trajectories (Fig 5(c)), where the open-loop approach frequently dips to low values with an average ERI of 38.6%, indicating that the reserve capacity is insufficient during specific periods. In contrast, ReserveMAX NMPC, on average, maintains an ERI of around 50%.

Table I shows the quantification performance of ReserveMax NMPC compared to the open-loop optimization approach. By leveraging interdependencies and closed-loop formulation, the operation of both systems is improved in terms of resilience and economy. Although economic dispatch formulation from open-loop is expected to provide lower operational costs, the sudden jumps in pump operation during peak water demands and frequent charging and discharging periods of the BESS cause substantial increases in the MG operation cost. Not only did the ReserveMAX NMPC offer lower operational costs, but it also increased the average ERI to enhance MG capacity in meeting the critical load, maintaining an optimal level of BESS, and better water reserve management to support preparedness for peak and emergency demand.

VI. CONCLUSION AND FUTURE WORK

This paper presents the combined operation of WDS and grid-connected MG, promoting the system's reliability. This

is done by utilizing ReserveMAX NMPC, which prioritizes maximizing power reserves from each generation unit while ensuring cost-efficient operation across both systems. The findings demonstrate the robustness of the proposed control strategy, which effectively adapts to seasonal variations in water demand and available renewable sources. This is evidenced by its continued compliance with water and power demands while maintaining an enhanced power reserve, averaging 1000kW. Furthermore, this study compared the ReserveMAX NMPC performance to an open-loop approach. Consistent improvement is obtained with a 30% increase in energy reserve, a 20% cost reduction, a 15% increase in energy storage, and a 20% improvement in water reserves. Future work will focus on expanding the model to incorporate larger networks and systems dynamics, enabling a more comprehensive evaluation of real-world implementations. Specifically, this includes incorporating 1) head-flow-based relationships into WDS dynamics to ensure effective pressure head management and 2) inverter-based distributed energy resources dynamics to ensure optimal active and reactive power management.

REFERENCES

- [1] R. O'Neil, K. Oikonomou, V. Tidwell, N. Voisin, J. Kerby, Z. J. Hou, M. Parvania, A. T. Al-Awami, M. Panteli, S. A. Conrad *et al.*, "Global research priorities for holistic integration of water and power systems," *IEEE Open Access Journal of Power and Energy*, 2024.
- [2] K. Oikonomou and M. Parvania, "Optimal coordination of water distribution energy flexibility with power systems operation," *IEEE Transactions on Smart Grid*, vol. 10, no. 1, pp. 1101–1110, 2018.
- [3] T. Shmaya and A. Ostfeld, "Conjunctive optimal design of water and power networks," *Journal of Hydrology*, vol. 643, p. 131932, 2024.
- [4] C. Wang, N. Gao, J. Wang, N. Jia, T. Bi, and K. Martin, "Robust operation of a water-energy nexus: A multi-energy perspective," *IEEE Transactions on Sustainable Energy*, vol. 11, no. 4, pp. 2698–2712, 2020.
- [5] F. Jalilian, M. A. Mirzaei, K. Zare, B. Mohammadi-Ivatloo, M. Marzband, and A. Anvari-Moghaddam, "Multi-energy microgrids: An optimal dispatch model for water-energy nexus," *Sustainable cities and society*, vol. 77, p. 103573, 2022.
- [6] F. Moazeni and J. Khazaei, "Optimal operation of water-energy microgrids; a mixed integer linear programming formulation," *Journal of Cleaner Production*, vol. 275, p. 122776, 2020.
- [7] M. Elsir, A. S. Al-Sumaiti, M. S. El Moursi, and A. T. Al-Awami, "Coordinating the day-ahead operation scheduling for demand response and water desalination plants in smart grid," *Applied Energy*, vol. 335, p. 120770, 2023.
- [8] A. Bhatraj, E. Salomons, and M. Housh, "An optimization model for simultaneous design and operation of renewable energy microgrids integrated with water supply systems," *Applied Energy*, vol. 361, p. 122879, 2024.
- [9] C. Bordons, F. Garcia-Torres, and M. A. Ridao, *Model predictive control of microgrids*. Springer, 2020, vol. 358.
- [10] S. A. Putri, F. Moazeni, and J. Khazaei, "Predictive control of interlinked water-energy microgrids," *Applied Energy*, vol. 347, p. 121455, 2023.
- [11] B. H. Vu and I.-Y. Chung, "Optimal generation scheduling and operating reserve management for pv generation using rnn-based forecasting models for stand-alone microgrids," *Renewable Energy*, vol. 195, pp. 1137–1154, 2022.
- [12] M. Quashie, C. Marnay, F. Bouffard, and G. Joós, "Optimal planning of microgrid power and operating reserve capacity," *Applied Energy*, vol. 210, pp. 1229–1236, 2018.
- [13] T. M. Walski, D. V. Chase, D. A. Savic, W. Grayman, S. Beckwith, and E. Koelle, *Advanced Water Distribution Modeling and Management*. Haestad Press, 2003.
- [14] M. Hijjo, F. Felgner, and G. Frey, "PV-battery-diesel microgrid design for buildings subject to severe power outages," in *2017 IEEE PES PowerAfrica*. Accra: IEEE, Jun. 2017, pp. 280–285. [Online]. Available: <https://ieeexplore.ieee.org/document/7991237/>
- [15] F. Moazeni and J. Khazaei, "Optimal energy management of water-energy networks via optimal placement of pumps-as-turbines and demand response through water storage tanks," *Applied Energy*, vol. 283, p. 116335, Feb. 2021. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0306261920317189>